Yocto Project and core layers overview



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Corrections, suggestions, contributions and translations are welcome!



The Yocto Project overview

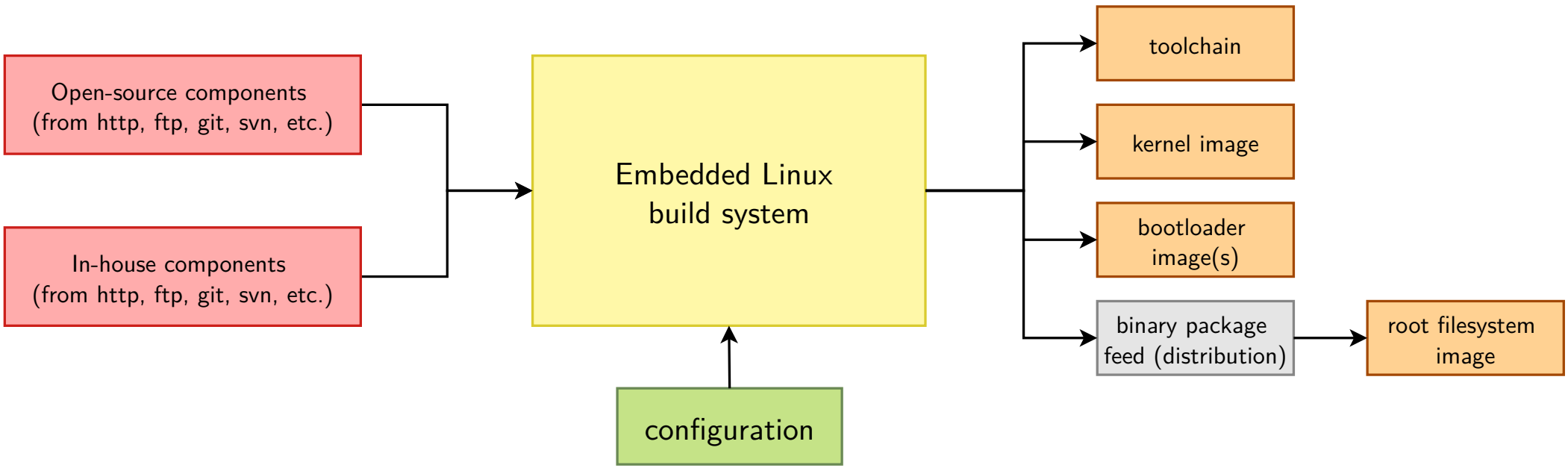


About

- The Yocto Project is an open source collaboration project that allows to build custom embedded Linux-based systems.
- Established by the Linux Foundation in 2010 and still managed by one of its fellows: Richard Purdie.



Yocto: principle



- Yocto always builds binary packages (a “distribution”)
- The final root filesystem is generated from the package feed
- The big picture is way more complex



Lexicon: `bitbake`

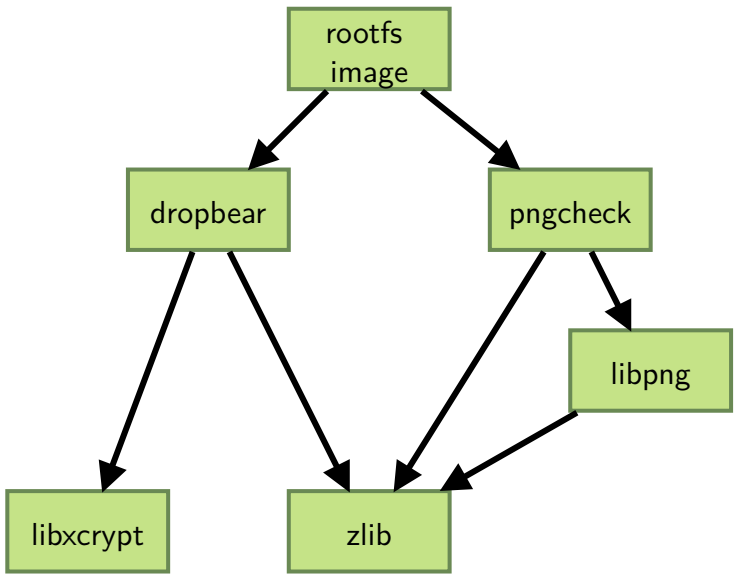
In Yocto / OpenEmbedded, the *build engine* is implemented by the `bitbake` program

- `bitbake` is a task scheduler, like `make`
- `bitbake` parses text files to know what it has to build and how
- It is written in Python (need Python 3 on the development host)



Lexicon: recipes

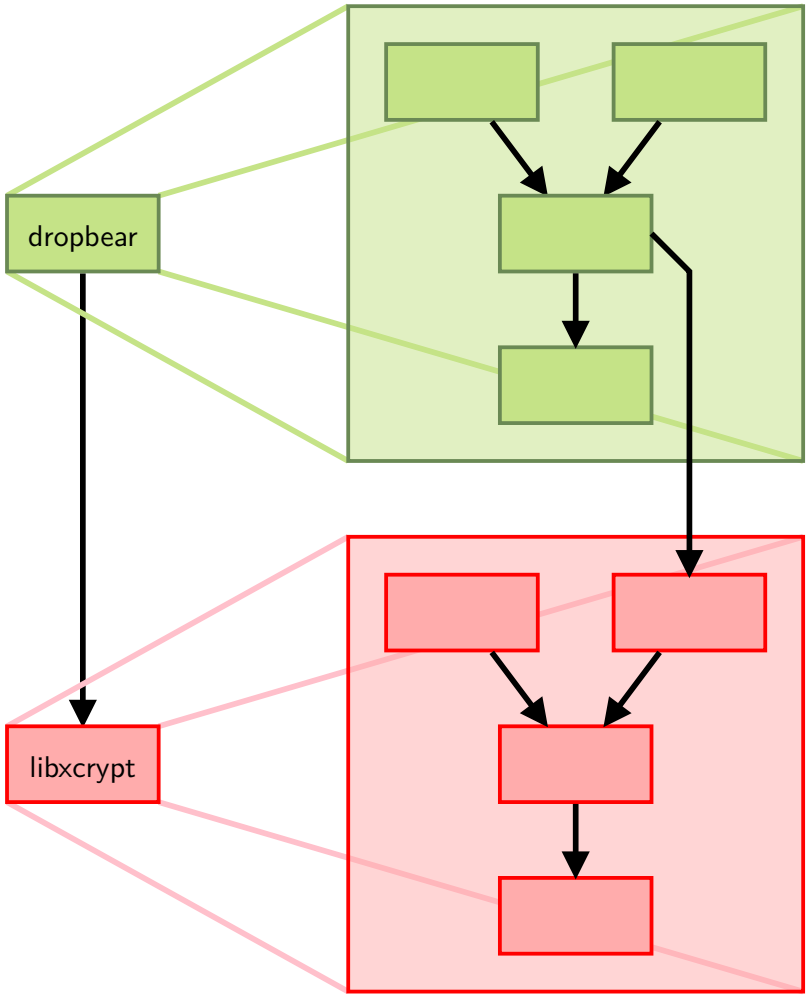
- The main kind of text file parsed by `bitbake` is *recipes*, each describing a specific software component
- Each *Recipe* describes how to fetch and build a software component: e.g. a program, a library or an image
- They have a specific syntax
- `bitbake` can be asked to build any recipe, building all its dependencies automatically beforehand





Lexicon: tasks

- The build process implemented by a recipe is split in several *tasks*
- Each task performs a specific step in the build
- Examples: fetch, configure, compile, package
- Tasks can depend on other tasks (including on tasks of other recipes)





Lexicon: metadata and layers

- The input to `bitbake` is collectively called *metadata*
- Metadata includes *configuration files*, *recipes*, *classes* and *include files*
- Metadata is organized in *layers*, which can be composed to get various components
  - A layer is a set of recipes, configurations files and classes matching a common purpose
    - For Texas Instruments board support, the *meta-ti-bsp* layer is used
  - Multiple layers are used for a project, depending on the needs
- *openembedded-core* is the core layer
  - All other layers are built on top of openembedded-core
  - It supports the ARM, MIPS (32 and 64 bits), PowerPC, RISC-V and x86 (32 and 64 bits) architectures
  - It supports QEMU emulated machines for these architectures

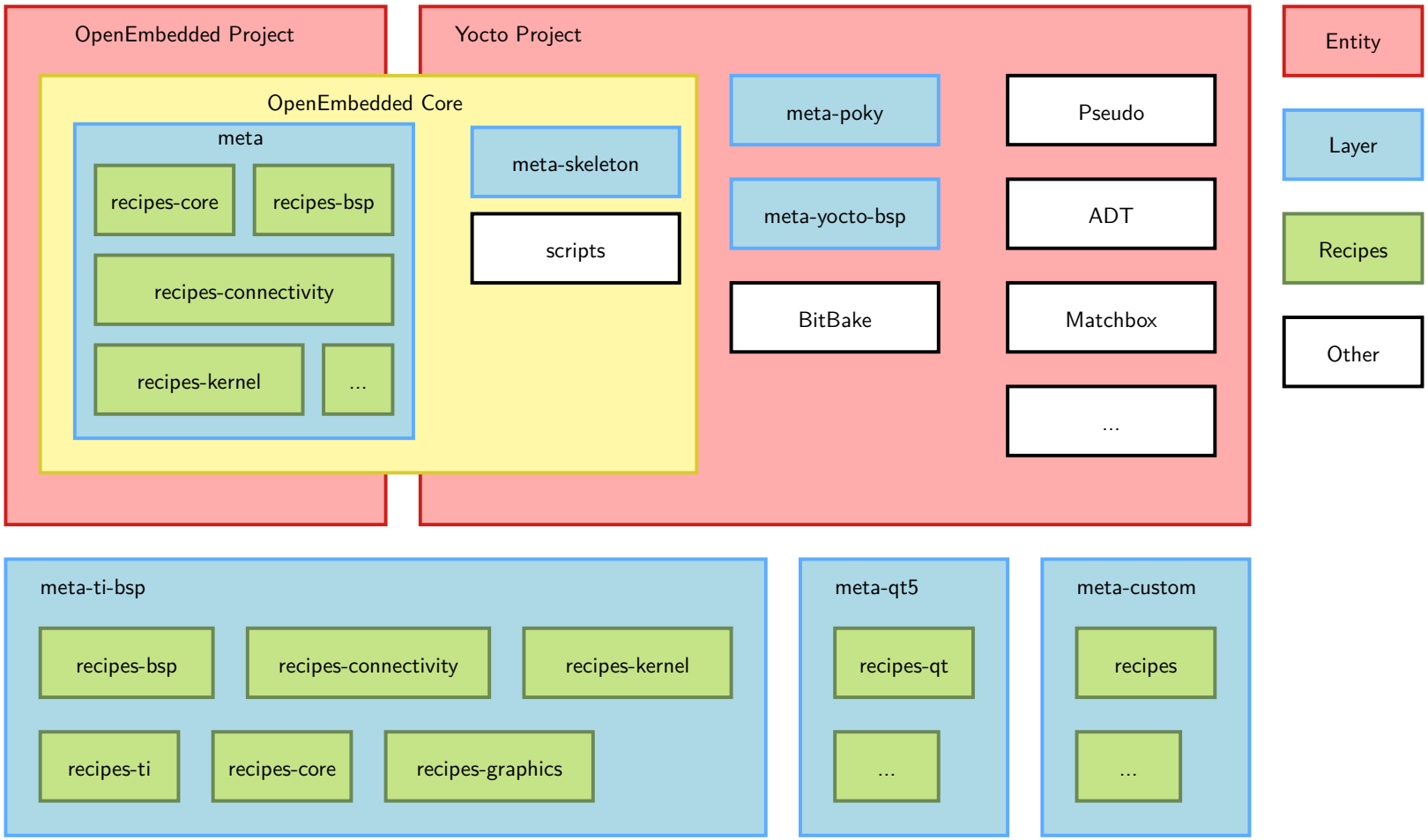


Lexicon: Poky

- The word *Poky* has several meanings
- poky is the *reference distro* provided by the Yocto Project
- meta-poky is the layer providing the poky reference distribution
- Up to the Walnascar (5.2) release, Poky was a git repository that was assembled from other git repositories: bitbake, openembedded-core, yocto-docs and meta-yocto



The Yocto Project lexicon





The Yocto Project lexicon

- The Yocto Project is **not used as** a finite set of layers and tools.
- Instead, it provides a **common base** of tools and layers on top of which custom and specific layers are added, depending on your target.



Example of a Yocto Project based BSP

- To build images for a BeagleBone Black, we need:
  - The Yocto core layers, containing all common recipes and tools.
  - The *meta-ti-bsp* layer, a set of Texas Instruments specific recipes.
- All modifications are made in your own layer. Editing openembedded-core or any other third-party layer is a **no-go**!
- We will set up this environment in the lab.



The Yocto core layers overview



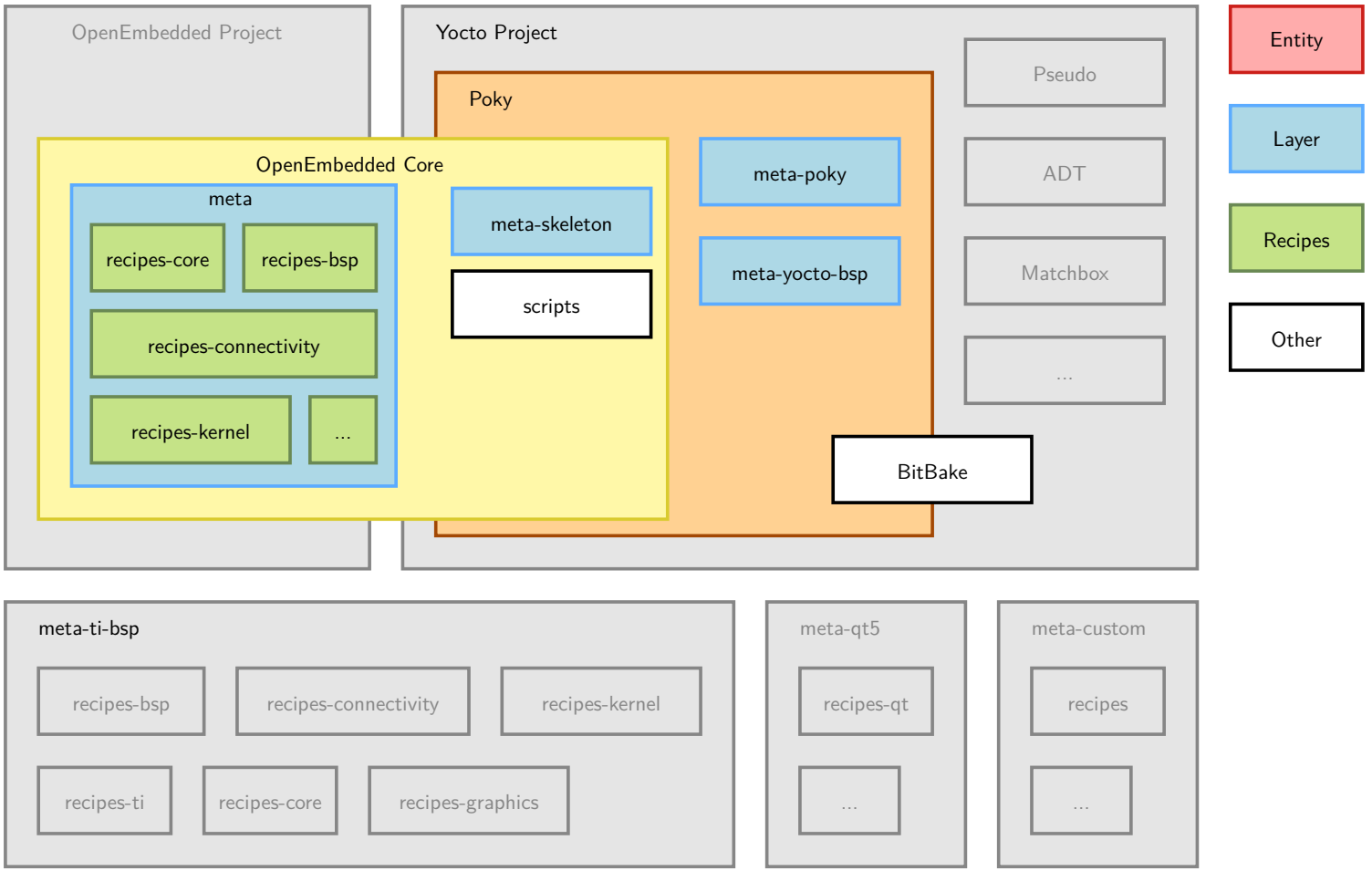
Getting the Poky reference system (up to Walnascar)

All official projects part of the Yocto Project are available at https://git.openembedded.org/ and https://git.yoctoproject.org/

- To download the Poky reference system:
  `git clone -b scarthgap https://git.yoctoproject.org/git/poky`
- A new version is released every 6 months, and maintained for 7 months
- **LTS** versions are maintained for 4 years, and announced before their release.
- Each release has a codename such as `kirkstone` or `scarthgap`, corresponding to a release number.
  - A summary can be found at https://www.yoctoproject.org/development/releases/



Poky


OpenEmbedded Project
Yocto Project
Poky
OpenEmbedded Core
meta
recipes-core
recipes-bsp
recipes-connectivity
recipes-kernel
...
meta-skeleton
scripts
meta-poky
meta-yocto-bsp
BitBake
Pseudo
ADT
Matchbox
...
Entity
Layer
Recipes
Other
meta-ti-bsp
recipes-bsp
recipes-connectivity
recipes-kernel
recipes-ti
recipes-core
recipes-graphics
meta-qt5
recipes-qt
...
meta-custom
recipes
...



Poky source tree

bitbake/	Holds all scripts used by the `bitbake` command. Usually matches the stable release of the BitBake project.
documentation/	All documentation sources for the Yocto Project documentation. Can be used to generate nice PDFs.
meta/	Contains the OpenEmbedded-Core metadata.
meta-skeleton/	Contains template recipes for BSP and kernel development.
meta-poky/	Holds the configuration for the Poky reference distribution.
meta-yocto-bsp/	Configuration for the Yocto Project reference hardware board support package.
LICENSE	The license under which Poky is distributed (a mix of GPLv2 and MIT).
oe-init-build-env	Script to set up the OpenEmbedded build environment. It will create the build directory.
scripts/	Contains scripts used to set up the environment, development tools, and tools to flash the generated images on the target.



Getting the Yocto reference layers (starting from Whinlatter)

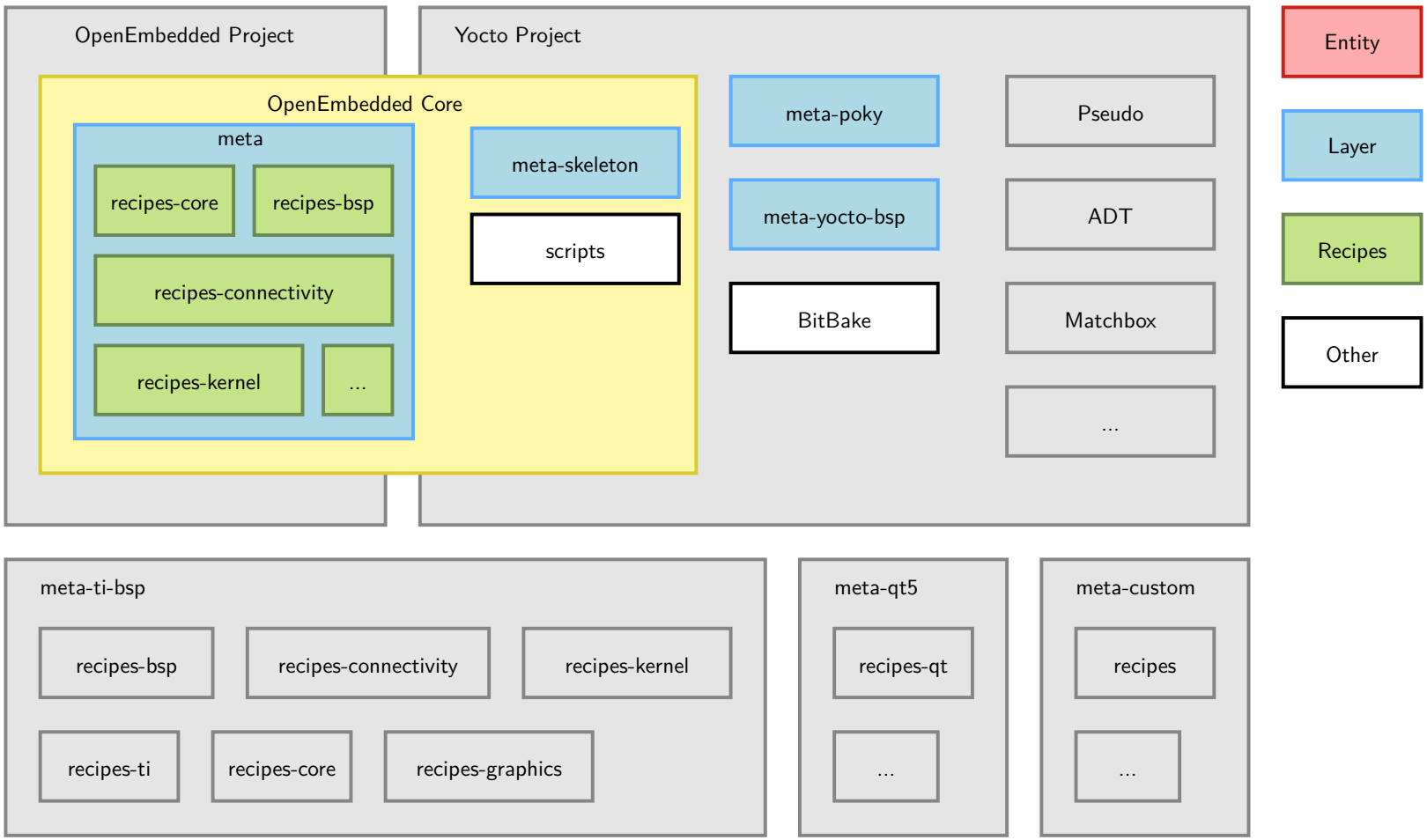
- The assembled poky git repository was deprecated with the Whinlatter (5.3) release.
- All source code is still the same but needs to be assembled from separate gits:
  - Either manually:

```
git clone https://git.openembedded.org/bitbake -b yocto-6.0
git clone https://git.openembedded.org/openembedded-core -b yocto-6.0
git clone https://git.yoctoproject.org/meta-yocto -b yocto-6.0
```

  - Or using the new bitbake-setup utility.
- Everything else stays the same.

```
poky/bitbake/                  bitbake/
poky/meta/                     openembedded-core/meta/
poky/oe-init-build-env    →    openembedded-core/oe-init-build-env
poky/meta-poky/                meta-yocto/meta-poky/
poky/meta-yocto-bsp/           meta-yocto/meta-yocto-bsp/
```

Getting the Yocto reference layers (starting from Whinlatter)





Yocto core layers source tree

bitbake/	Holds all scripts used by the `bitbake` command. Usually matches the stable release of the BitBake project.
openembedded-core/meta/	Contains the OpenEmbedded-Core metadata.
openembedded-core/meta-skeleton/	Contains template recipes for BSP and kernel development.
openembedded-core/LICENSE	The license under which openembedded-core is distributed (a mix of GPLv2 and MIT).
openembedded-core/oe-init-build-env	Script to set up the OpenEmbedded build environment. It will create the build directory.
openembedded-core/scripts/	Contains scripts used to set up the environment, development tools, and tools to flash the generated images on the target.
meta-yocto/meta-poky/	Holds the configuration for the Poky reference distribution.
meta-yocto/meta-yocto-bsp/	Configuration for the Yocto Project reference hardware board support package.



Documentation

- Documentation for the current sources, compiled as a “mega manual”, is available at: https://docs.yoctoproject.org/singleindex.html
- Variables in particular are described in the variable glossary: https://docs.yoctoproject.org/genindex.html